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Supplementary Material for Recovery-Aware Physical Attribution of Load and Line Events From Sparse PMUs

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ABSTRACT This supplement documents the implemented recovery-aware history bank: its local filtering calculations, measurement map, publication rules and simulation contracts. It records the complete confirmatory outcomes, controlled severity-score/state-carry ablation and continuous single-state control. The operating-point signature and recovery-geometry studies keep their separate, known-onset contracts and denominators. All empirical results reuse the fixed evidence; no new campaign or threshold selection is introduced here. Earlier voltage-reconstruction, simultaneous-source and general diagnosability studies are preserved in the separately compiled Historical Technical Notes.

INDEX TERMS phasor measurement units, sequential diagnosis, physical state continuity, conditional filtering, reproducibility.

APPENDIX A IMPLEMENTED PREDICTOR AND CONDITIONAL UNCERTAINTY

This section specifies the local operators and conditional distributions used by the fixed continuous estimator. It distinguishes identities of that local model from approximations introduced by its finite history bank. The numerical checks below exercise these identities on controlled inputs; they do not recalibrate or rerun the reported evaluation.

A. AFFINE OPERATORS AT A CONSISTENT TOPOLOGICAL ANCHOR

For network hypothesis ℓ, let $(\bar{x}_\ell, \bar{u}_\ell)$ be the operating point determined by the prefix-fitted load and dispatch factors. Fix its differential coordinates while solving the algebraic constraints separately for each topology q:

$$g_{\ell,q}(\bar{x}_\ell, \bar{z}_{\ell,q}, \bar{u}_\ell) = 0. \tag{1}$$

The resulting point need not be a differential equilibrium after a line opening. Assume a nonsingular algebraic Jacobian and a fixed angle gauge. In the expressions below, f is the differential vector field after division by any nonunit differential mass entries. All derivatives are evaluated at the topology-specific point in (1); the network index is suppressed for readability. Define

$$\begin{aligned} c_q &= f_q(\bar{x}, \bar{z}_q, \bar{u}), \\ A_q^c &= f_x - f_z g_z^{-1} g_x, & B_q^c &= f_u - f_z g_z^{-1} g_u, \\ C_q^c &= H_x - H_z g_z^{-1} g_x, & D_q^c &= H_u - H_z g_z^{-1} g_u, \\ d_q^{el} &= H_q(\bar{x}, \bar{z}_q, \bar{u}) - H_0(\bar{x}, \bar{z}_0, \bar{u}). \end{aligned} \tag{2}$$

For the admitted scalar load direction $\delta u = La$, the continuous approximation is $\dot{\delta x} = A_q^c \delta x + c_q + B_q^c La$ and $\delta y = C_q^c \delta x + d_q^{el} + D_q^c La$. For a line, the topology selects the operator and the amplitude direction is zero. A nonzero pre-event state contributes through δx; d_q^{el} supplies the zero-deviation offset at the fitted anchor. This is a local expansion, so large excursions can still cause prediction error even when the algebraic solve and derivatives are accurate.

For a mode lasting h, compute the affine flow without inverting A_q^c:

$$\mathcal{A}_q = \begin{pmatrix} A_q^c & c_q & B_q^c L \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$
$$\exp(h\mathcal{A}_q) = \begin{pmatrix} F_q(h) & \beta_q(h) & U_q(h) \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{3}$$

Thus $\beta_q(h) = \int_0^h \exp(A_q^c s) c_q \, ds$. If an admitted switch divides one report into an old-mode interval h_1 and a new-mode interval h_2, then

$$F = F_2 F_1, \qquad \beta = F_2 \beta_1 + \beta_2, \qquad U = F_2 U_1 + U_2. \quad (4)$$

The physical state is continuous across this composition. The final-mode measurement map uses the consistent algebraic right limit. The predictor then appends phase history and the deterministic memory of the instrumental transformation. Scaling these coordinates and applying the same affine augmentation to (4) produces the stored $T_o, J_o, g_o, u_o, j_o, v_o$ operators. In their report ordering,

$$\begin{aligned} r_k &= J_o \xi_k + j_o + v_o a + \epsilon_k, \\ \xi_{k+1} &= T_o \xi_k + g_o + u_o a + \eta_k. \end{aligned} \quad (5)$$

The measurement is scored before correcting the state for the next report. The constant startup discrepancy is applied to the observations; it does not replace the topology-dependent affine terms.

The construction has three steps. A static solve of (1) exports the network forcing, the PMU reference and their reduced Jacobians; the conditional builder forms the topology offsets, the mode flows and the one-report operators; and a final check verifies that the constructed reference matches the prefix-fitted operating point. The reproduction manifest preserves these files and their hashes. No event truth is used to select the startup operating point.

B. FOUR AMPLITUDE INTERVALS REPRESENT ONE PRIOR

Let $I = [-0.2, 0.2]$ be partitioned into four disjoint intervals I_j up to endpoints of zero probability. Write $Z_J(\mu, v) = \int_J \mathcal{N}(a; \mu, v)\, da$ for any interval J. The initial prior has the exact decomposition

$$\begin{aligned} p_0(a) &= \sum_{j=1}^{4} \alpha_{j,0} \, \mathrm{TN}_{I_j}(a; \mu_0, v_0), \\ \alpha_{j,0} &= Z_{I_j}(\mu_0, v_0) / Z_I(\mu_0, v_0). \end{aligned} \quad (6)$$

The implementation assigns one history per source, proposed onset, restoration path, and interval. Each retains its own $(m_{h,j}, d_{h,j}, \mu_{h,j}, v_{h,j}, w_{h,j})$; there is no per-report collapse to a single amplitude distribution. The four interval histories use the same physical operators. They are neither quadrature nodes nor four separately linearized severity models.

For a fixed physical history with observations Y, let $Z_j(Y) = \int_{I_j} p(Y \mid a, h)\, \mathrm{TN}_{I_j}(a; \mu_0, v_0)\, da$. Then

$$p(Y \mid h) = \sum_j \alpha_{j,0} Z_j(Y), \qquad \alpha_j(Y) = \frac{\alpha_{j,0} Z_j(Y)}{p(Y \mid h)}. \quad (7)$$

Applying the local predictive integral sequentially gives the same identity. Identical linear operators and gains preserve the common untruncated Gaussian parameters for interval copies of the same path; their truncated moments and normalized weights differ. Subsequent beam pruning can remove interval histories, so equivalence with an unpartitioned bank is not guaranteed after pruning. The recorded discarded mass describes that approximation and is not a bound on all future decisions.

C. AMPLITUDE DEPENDENCE AND MIXTURE COMPRESSION

Condition on a retained history and the reports used so far. For $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, set $\sigma = \sqrt{v}$, $\alpha = (l-\mu)/\sigma$, $\beta = (u-\mu)/\sigma$, $Z = \Phi(\beta) - \Phi(\alpha)$, and $\lambda = [\phi(\alpha) - \phi(\beta)]/Z$. Its moments are

$$\begin{aligned} \bar{a} &= \mu + \sigma\lambda, \\ V_a &= v \left[1 + \frac{\alpha\phi(\alpha) - \beta\phi(\beta)}{Z} - \lambda^2 \right]. \end{aligned} \quad (8)$$

The truncated-moment routine evaluates the normalizer in the log domain and bounds small numerical excursions of the moments. The Gaussian parameters μ, v are therefore distinct from the conditional amplitude mean and variance.

For the affine conditional state model $\xi \mid a, h \sim \mathcal{N}(m_h + d_h a, P_h)$, total expectation and covariance give

$$\begin{aligned} \bar{\xi}_h &= m_h + d_h \bar{a}_h, \\ \mathrm{Cov}(\xi, a \mid h) &= d_h V_{a,h}, \\ V_h &= P_h + d_h V_{a,h} d_h^\top. \end{aligned} \quad (9)$$

With normalized history weights ω_h, including the no-new-event history, the complete mixture moments would be

$$\begin{aligned} \bar{\xi} &= \sum_h \omega_h \bar{\xi}_h, \\ P_{\mathrm{mix}} &= \sum_h \omega_h \left[V_h + (\bar{\xi}_h - \bar{\xi})(\bar{\xi}_h - \bar{\xi})^\top \right]. \end{aligned} \quad (10)$$

This identity applies within a network hypothesis and its common state coordinates. It does not justify averaging absolute rotor states between different network equilibria.

The continuous estimator uses the shared regime covariance P_ρ in place of P_h. It updates the affine means and amplitude distributions, and computes marginal diagonal state variances for diagnostics. At closure it carries $\bar{\xi}$, clears explicit histories, and records the trace of amplitude/mean spread omitted from the new single-history representation. It does not carry the complete covariance in (10), nor the uncertainty of the fitted startup context. Instrumental and discrepancy states remain in the carried augmented mean.

For comparison, an exact conditional covariance recursion for the local linear model (5), with independent noises of covariances R, Q, is

$$\begin{aligned} S_h &= J_o P_h J_o^\top + R, \\ K_h &= T_o P_h J_o^\top S_h^{-1}, \\ P_h^+ &= T_o P_h T_o^\top + Q - K_h S_h K_h^\top. \end{aligned} \quad (11)$$

This is a diagnostic reference, not a description of a new executed continuous estimator. The fixed implementation solves the Riccati equation for the nominal operator in each

TABLE 1. Deterministic local-model checks. Maximum absolute errors over 72 scalar updates, five switch positions, and 12 joint Gaussian updates; each row uses its stated reference.

Check	Maximum error
Log evidence: partition	1.78×10^{-15}
Amplitude mean/variance	4.16×10^{-17}
Log evidence: quadrature	3.11×10^{-15}
Affine flow: ODE reference	2.22×10^{-16}
Joint mean/covariance	5.55×10^{-17}

discrepancy regime and uses that shared covariance for the admitted switched operators.

Sequential likelihood factorization conditions on past reports; it does not require marginal independence of the observations. The instrumental VAR/AR filter and the latent discrepancy AR state already model temporal dependence. Remaining covariance misspecification, linearization error, adaptive proposals, and compression can nevertheless affect the resulting weights. Their empirical calibration requires pre-update residual and coverage diagnostics, separately from the algebraic checks below.

D. DETERMINISTIC VERIFICATION AND ITS BOUNDARY

A deterministic check loads the implemented amplitude routine without running a campaign. It compares sequential one-interval and four-interval evidence and moments with direct scalar quadrature, including zero information and amplitudes on interval boundaries. An independent small switched linear system checks affine-flow composition against numerical integration. A linear Gaussian reference also compares the conditional affine-state recursion with a joint state–amplitude Kalman recursion for an untruncated Gaussian amplitude; the truncated case is checked separately by quadrature. All histories are retained in these checks. Source hashes, tolerances, and complete results accompany the generated table. These checks establish local-model identities on the stated inputs. They establish neither calibration of the pruned network bank nor robustness of its nonlinear physical approximation, and add no event outcomes to the evaluation cohorts.

APPENDIX B HOW TO READ AND REPRODUCE THE CURRENT METHOD

This companion begins with the recovery-aware causal method used in the current article. Sections B–D give its execution order, local derivations, worked examples and simulation contracts. The separately compiled Historical Technical Notes preserve full diagnosability proofs, earlier state-reconstruction and batch-estimator studies, and secondary evidence. The earlier studies support particular components; they are not additional tests of the complete causal algorithm.

The article retains the physical model, the defining evidence integral, the recovery mechanism, the decision outputs and decisive results. This supplement supplies the intermediate calculations. The Historical Technical Notes develop functional observability, nuisance-profiled information and source ambiguity, and distinguish the implemented batch estimator $\mathcal{E}_{\mathrm{B}}$ from the proposed joint estimator $\mathcal{E}_{\mathrm{J}}$. Neither should be silently substituted for the causal physical history bank. The conditional Gaussian updates use standard filtering identities [1]; the derivation below specifies their role and the approximations made by the implemented bank.

A. INPUTS, OUTPUTS AND THE INFORMATION BOUNDARY

The inputs are successive reports from eight PMUs, the fixed physical and instrumental operators, the admitted load/line catalog and the fixed method configuration. The output is a timestamped sequence of detection, family/source and restoration messages, together with retained attribution masses. True ON/OFF times, true sources and generating severities belong to the evaluator. They are not inputs to the causal algorithm.

Three times must remain distinct: the latest observed report, the estimated physical onset, and the public emission time. Looking back to propose an onset is compatible with causality when the message is emitted only after the required reports arrive. It does not permit backdating the public alarm.

B. EXECUTION ORDER FOR ONE STREAM

1) **Load the fixed operators.** Retain voltage reference, oriented terminal currents, load input directions, switch modes and instrument memory. A line opening changes the network and observation operators. It is not represented by fitting a load amplitude.
2) **Fit the startup context.** Use the first 300 reports for operating factors, common reference and noise scale. For each network hypothesis, set the voltage/current discrepancy to the difference between the observed prefix mean and its reference prediction. Keep that offset fixed afterwards. Frequency and ROCOF use their empirical prefix centers. The prefix is not scored again as independent model-selection evidence.
3) **Predict the next report.** Propagate the retained histories and evaluate each network model's predictive density before correcting states or choosing new histories with the current report.
4) **Propose candidate onsets.** The fixed screen uses lags 0–18 reports within a ten-second stored state history. Each proposal starts from a stored state before its proposed onset, including any recovery already present. The bank does not enumerate every possible onset in the stored interval.
5) **Score histories.** For loads, integrate severity with the bounded prior and the local conditional Gaussian likelihood. For lines, score the admitted switched operator. Score a report before using that same report for its conditional state correction.
6) **Update and compress.** Update severity and conditional states, network weights, restoration children and the retained beam. Record discarded mass. At episode closure retain the augmented-state mixture mean; the

omitted between-history spread is a documented approximation, not an exact posterior covariance.

7) **Publish and record.** Aggregate active mass M, cause mass Γ and ΔM. Apply the fixed detection, diagnosis and restoration rules. Save emission time and the last report used, in addition to estimated onset and source. Continue propagation through OFF.

C. WHY REPLAYED REPORTS ARE NOT NEW EVIDENCE

Let a retained parent have log score $\ell_{0:k} = \ell_{0:\tau-1} + \ell^{old}_{\tau:k}$. Replacing its suffix by a newly proposed physical history gives

$$\ell^{new}_{0:k} = \ell_{0:k} - \ell^{old}_{\tau:k} + \ell^{new}_{\tau:k}. \tag{12}$$

Transition and prior terms must accompany their corresponding history. The rolling screen supplies a proposal, not an extra independent likelihood. Adding the new suffix without removing the old suffix would count the same reports twice. This bookkeeping statement does not make adaptive screening or beam pruning exact Bayesian inference.

D. FIXED NUMERICAL CHOICES

The input grid is 30 reports/s. The five network hypotheses are nominal, global $\pm 2\%$ and selected-branch $\pm 3\%$ paired R/X perturbations. Each hypothesis is fixed in time; the weights change. They do not identify a continuous impedance posterior.

The per-report birth and restoration hazards are $1 - \exp[-(1/60)/30]$ and $1-\exp[-(1/30)/30]$. Birth mass is divided equally between load and line families, then uniformly among the admitted sources within each family. Load severity has a zero-mean Gaussian prior with standard deviation 0.15, truncated to $[-0.2, 0.2]$ and integrated over intervals bounded by $-0.2, -0.1, 0, 0.1, 0.2$. After three reports, eligible active histories can create OFF children at either half-report position, inheriting state and severity. After likelihood normalization, the beam retains the union of the 96 highest-weight recent histories, 64 older histories and at least one per occupied source. The union is renormalized.

The internal observer confirms active-mass evidence at 0.99 for six reports and has a separate 15-report internal OFF clock. The public physical-bank publisher requires an internal pending episode and six reports of source-qualified active support above 0.99; its public OFF requires six reports of the diagnosed cell's joint cause-and-restoration mass above 0.99. The detailed source qualification, diagnosis and replacement rules are given below. These thresholds are implementation settings, not guarantees of probability calibration or false-alarm control. Nominal and mixture use their fixed publication paths, so their comparison does not isolate network averaging alone. The admission-rule ablation changes publication, not the recovery dynamics.

E. SOURCE QUALIFICATION AND PUBLIC EPISODE RULES

The source screen and the onset search are distinct. Source qualification uses a 30-report (one-second) conditional innovation vector v with 47 coordinates per report. Fixed normal-mode gains propagate the stored pre-window state and whiten the suffix. For each prepared response s_j and lag from 0 to 25 reports, let

$$\begin{aligned} g_j &= s_j^\top s_j, \qquad u_j = s_j^\top v, \qquad R = v^\top v, \\ \widehat{a}_j &= \mathrm{clip}(u_j / \max(g_j, 10^{-30}), -0.2, 0.2), \\ G_j &= 2\widehat{a}_j u_j - \widehat{a}_j^2 g_j, \\ F_j &= \frac{\max(G_j, 0)}{\max\{(R - G_j)/1409, 1\}}. \end{aligned} \tag{13}$$

Line columns use fixed amplitude one. The source score is the maximum over its columns; the threshold is the $1 - 0.001/936$ quantile of an $F_{1,1409}$ distribution, approximately 24.0115. This threshold and the variance floor are a fixed proposal/publication rule, not a calibrated global false-alarm probability. The statistic is never multiplied into the Bayesian history likelihood. This one-second qualification screen does not enlarge the separate birth search beyond its 0–18 report lags.

Define $b_c = 1$ for a line, and $b_c = \mathbf{1}\{F_c > 24.0115\}$ for a load. At each monitored report, the publisher updates consecutive-report counters for $M_c b_c \geq 0.99$, $\Gamma_c \geq 0.99$, and $\Gamma_c P(\mathrm{OFF} \mid c) \geq 0.99$. A false condition clears its counter. An internal alarm opens a pending episode. A public alarm then requires either six pending reports with $\sum_c M_c b_c \geq 0.99$ or an already accumulated six-report line-active counter. The latter permits line evidence collected before the internal pending message to count.

A public episode receives one diagnosis when its highest cause-mass cell has six consecutive reports above 0.99, or at the 15-report deadline after the public alarm. The output family maximizes summed cause mass; sources are ranked within that family. The diagnosed source becomes the episode owner. Six reports of that owner's joint cause-and-OFF mass above 0.99 emit a public restoration. These publication operations do not rescale M or reset the physical state.

The internal mixture lifecycle separately replaces an episode after six reports with mass at least 0.99 on active histories whose onset is later than its alarm (using the implementation's quarter-report timing guard). It withdraws an unconfirmed episode after the diagnosis delay when null mass has exceeded 0.99 for 15 reports. Its internal restoration requires cause mass at least 0.99 and conditional retired mass at least 0.99 for 15 reports. These terminal messages are propagated to the public episode if already emitted; an unconfirmed pending episode is recorded as rejected. Finalization marks any remaining public episode right-censored. The nominal comparator retains its own fixed internal lifecycle, so the nominal/mixture comparison is not an isolated model-averaging ablation.

APPENDIX C LOCAL CALCULATIONS AND WORKED EXAMPLES

A. COMPLETING THE BOUNDED-AMPLITUDE INTEGRAL

For one retained severity interval, write $e = sa + \epsilon$, with $\epsilon \sim \mathcal{N}(0, S)$, $S \succ 0$, and $a \sim \mathrm{TN}_{[l,u]}(\mu, v)$, $v > 0$. Here μ, v are parameters of the underlying untruncated Gaussian, not the bounded distribution's moments. With $g = s^\top S^{-1} s$ and $q = s^\top S^{-1} e$, expansion gives

$$\begin{aligned}(e - sa)^\top S^{-1}(e - sa) + (a - \mu)^2/v \\ = e^\top S^{-1} e + \mu^2/v - \mu'^2/v' + (a - \mu')^2/v', \qquad (14)\\ v' = (v^{-1} + g)^{-1}, \qquad \mu' = v'(\mu/v + q). \qquad (15)\end{aligned}$$

Define $Z(\mu, v) = \Phi((u-\mu)/\sqrt{v}) - \Phi((l-\mu)/\sqrt{v})$. Integrating the last Gaussian square over $[l, u]$ yields

$$\begin{aligned}\log L = & -\tfrac{1}{2}[e^\top S^{-1} e + \log|S| + n_y \log(2\pi)] \\ & + \tfrac{1}{2}[\log(v'/v) + \mu'^2/v' - \mu^2/v] \\ & + \log Z(\mu', v') - \log Z(\mu, v). \qquad (16)\end{aligned}$$

The posterior on that interval is $\mathrm{TN}_{[l,u]}(\mu', v')$. With $\alpha = (l - \mu')/\sqrt{v'}$, $\beta = (u - \mu')/\sqrt{v'}$, $Z' = \Phi(\beta) - \Phi(\alpha)$ and standard normal density ϕ, its mean is

$$\mathbb{E}[a \mid e] = \mu' + \sqrt{v'}\,\frac{\phi(\alpha) - \phi(\beta)}{Z'}. \qquad (17)$$

Thus neither the untruncated mean μ' nor a best-fit severity can simply replace the integral. The implementation combines retained intervals using their weights. The identity is exact for this conditional Gaussian model; it does not remove linearization, shared-covariance or pruning error.

B. NUMERICAL EXAMPLE OF ONE INTERVAL

This dimensionless example illustrates the calculation; it is not an IEEE-39 simulation or a new performance result. Set $S = 1$, $s = 2$, $e = 1$, $\mu = 0$, $v = 1$ and $[l, u] = [0, 1]$. Then $g = 4$, $q = 2$, $v' = 0.2$ and $\mu' = 0.4$. Substitution in (16) gives $L \simeq 0.342688$ and $\mathbb{E}[a \mid e] \simeq 0.464941$. The different posterior mean follows from truncation. A separate numerical check compares the closed form against direct numerical integration, including the posterior mean. It checks algebra only and does not modify a fixed experiment.

C. CONDITIONAL STATE CORRECTION

Represent the conditional state mean as $m_h + d_h a$. Let $T_o, J_o, g_o, u_o, j_o, v_o$ denote the stored one-report transition and observation operators for switch o. The code forms $e = r_k - J_o m_h - j_o$ and $s = J_o d_h + v_o$. With shared covariance P_ρ in discrepancy regime ρ,

$$\begin{aligned}S_o = R + J_o P_\rho J_o^\top, \qquad K_o = T_o P_\rho J_o^\top S_o^{-1}, \\ m_h^+ = T_o m_h + g_o + K_o e, \quad d_h^+ = T_o d_h + u_o - K_o s. \qquad (18)\end{aligned}$$

To see the last line, apply the correction to a fixed amplitude:

$$T_o(m_h + d_h a) + g_o + u_o a + K_o(e - sa) = m_h^+ + d_h^+ a.$$

The severity distribution is updated by the same likelihood integral. Regime covariances come from nominal-operator Riccati equations. The physical history bank does not propagate an independent exact covariance for every switched history. At closure the mean is retained and the omitted mixture spread recorded.

D. RESTORATION IS NOT A STATE RESET

Consider the illustrative scalar system $\dot{x} = -x + u$, $y = x$, with $x(0) = 0$. Apply $u = a$ for $0 \le t < \kappa$ and restore $u = 0$ afterwards. Direct integration gives

$$x(t) = \begin{cases} a(1 - e^{-t}), & 0 \le t < \kappa, \\ a(1 - e^{-\kappa})e^{-(t-\kappa)}, & t \ge \kappa. \end{cases} \qquad (19)$$

Both expressions agree at κ. For $a = 1$ and $\kappa = 2$, $x(2) = 0.864665$ and $x(3) = 0.318092$ even though the intervention is OFF. Resetting the reference to equilibrium would discard that predictable recovery. This example explains state continuity; it does not claim that one exponential describes the multimode IEEE-39 response.

E. ACTIVE MASS, CAUSE MASS AND MODEL REWEIGHTING

For one model, suppose an illustrative cell has mass 0.60 in active histories and 0.30 in restored histories, with 0.10 elsewhere. Its active mass is $M_c = 0.60$, while its originating-cause mass is $\Gamma_c = 0.90$. Recovery can therefore remain attributable without declaring the input active. These assigned masses illustrate definitions, not calibrated probabilities from an experiment.

For two reports, adding and subtracting $\sum_\ell w_{\ell,k} M_{\ell,c,k-1}$ proves

$$\begin{aligned}\Delta M_c = & \sum_\ell w_{\ell,k}(M_{\ell,c,k} - M_{\ell,c,k-1}) \\ & + \sum_\ell (w_{\ell,k} - w_{\ell,k-1}) M_{\ell,c,k-1}. \qquad (20)\end{aligned}$$

For example, unchanged conditional masses $(0.8, 0.2)$ and network weights changing from $(0.5, 0.5)$ to $(0.75, 0.25)$ move M_c from 0.50 to 0.65. The entire 0.15 change comes from model reweighting. A rising aggregate mass need not mean that each model found stronger within-model evidence.

APPENDIX D SIMULATION SCENARIOS AND REPRODUCTION ORDER

A. PLANT, MEASUREMENT AND PROVENANCE

The simulator is a Julia/PowerDynamics research model of the IEEE 39-bus system, not an officially validated equivalent of the competition reference record. Its contract fixes Julia 1.11.9, PowerDynamics 5.0.0, NetworkDynamics 1.3.0 and IEEE39 0.1.0. Initial states, effective parameter vectors, terminal map, reporting operator and stochastic model come from that snapshot. Nominal inventory files do not replace the effective operating point. The load law is the fixed impedance law.

The voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. Each current is an oriented branch-terminal measurement, not a

nodal injection. BUS5 retains the provisional historical 5–6 branch, terminal 5, hypothesis; producer CT identity is unconfirmed and BUS39 remains partially validated. Execution uses full-precision configuration files rather than rounded tables. The competition reference record provided retrospective calibration context only.

Each physical parent is integrated continuously through its interventions. Differential states remain continuous at switching, while algebraic values are recomputed consistently. Reports use $t = k/30$ with endpoints included; the switch-aligned report uses the post-switch algebraic value. Solver steps are recorded separately and observations are not interpolated across a switch. The causal instrumental history is preserved through OFF.

B. CONTINUOUS DEVELOPMENT AND CONFIRMATION

The two development campaigns each contain three 80-s event parents with six interventions and one 80-s normal parent. Crossing the four parents with three R/X conditions gives 12 physical trajectories, 18 unique interventions and 54 event realizations per campaign. Both are development evidence in the current article. Repeated conditions are not independent event draws.

The confirmatory evaluation prespecifies 18 continuous parents, 27 persistent-line parents (nine lines, three repetitions), and 12 normal parents. The three R/X conditions produce $3(18+27+12) = 171$ intended physical trajectories; two instrumental-noise scales yield 342 observed records. The continuous parents contain $18 \times 6 = 108$ unique interventions. Their six condition/noise realizations are repeated observations of those parents. The 120-s and 600-s grids contain 3601 and 18001 reports, respectively. Completed results appear in the dedicated confirmation tables.

Continuous evaluation pairs an alarm temporally before inspecting its predicted label. The first unmatched post-ON alarm is eligible up to the next ON, with the inherited 30-s cap. Timely attribution requires correct family and asset emitted before physical OFF. Unmatched alarms, false restorations and open episodes are retained. Persistent-line evaluation instead scores the first stopping decision, including early stops, wrong sources, omissions and censoring. The detector of Chen *et al.* is evaluated only there; it has no load or restoration output. Confidence intervals use 10,000 paired parent-cluster resamples, not independent PMU samples.

C. SEPARATE SIGNATURE AND RECOVERY CONTROLS

The context-signature study uses a single load at buses $\{3, 7, 16, 24\}$, a 10-s record, known ON at 3 s and OFF at 6 s, and one scalar operating-context direction. It contains 57 dictionary/check trajectories and 85 test trajectories (80 interventions and five controls), with three common noise seeds producing 255 observed test records. The context and severity are estimated from measurements, while onset and family are supplied. The generating source is used to assess the response approximation and severity error. Those conditional metrics do not measure unknown-onset causal attribution.

The recovery-geometry control uses 38 trajectories. Prior load histories at buses 7 and 24, followed by delays of 0.5 and 2 s, give four recovery contexts. Three source-pair comparisons within each context give 12 correlated comparisons. The prior input is active from 2 to 3 s; the subsequent response is examined over 1.5 s. The supplied history, onset, family and two-candidate universe define an oracle control, with one uncertain prior-severity direction. It does not evaluate the causal physical method, unknown onset, event cardinality or operational false alarms.

D. REPRODUCTION WITHOUT CHANGING THE EXPERIMENT

1) Read each campaign's fixed plan and manifest. Select the intended population before interpreting any denominator. Keep current causal, signature, recovery-control and historical batch studies separate.
2) Resolve the plant state, parameter and measurement files from the versioned snapshot. Every new simulation is a new run, and a new hypothesis a new candidate; earlier preparation steps are never rerun inside a fixed baseline.
3) Preserve ON/OFF continuity, report endpoints, noise seeds and the boundary between observations and reserved event truth. Run inference before releasing truth to the evaluator.
4) Retain solver/inference failures in intended denominators. Aggregate at the declared parent level and preserve the original comparators and publication rules. Do not retune from confirmation results.
5) Rebuild the two documents from the versioned tables and figures. The default build does not regenerate scientific artifacts or import new campaign results; refreshing artifacts is a separate operation.

The reproducibility package keeps this walkthrough, the campaign evidence and the earlier mathematical development, compiled separately as Historical Technical Notes. A successful PDF build verifies document consistency; it is not a new scientific validation.

APPENDIX E EXECUTED MEASUREMENT MAP

A. MEASUREMENT CONTRACT

The executed channel map is taken from the versioned research-plant snapshot. The eight voltage buses are $\{2, 5, 6, 10, 19, 22, 29, 39\}$. The internal voltage phasor is in per unit with unit gain and is exported as phase-to-neutral RMS volts using 345 kV/$\sqrt{3}$; voltage angle uses one common affine phase reference plus the stored relative offsets. The reference interpretation is observational: physical common drift and instrument clock drift are not separately identified. Terminal currents are converted to outward branch convention from the simulator's into-terminal observable, then mapped with the row-specific polarity, gain and phase bias. Current bases use $100\,\mathrm{MVA}/(\sqrt{3}\,V_{\mathrm{terminal}})$; the common voltage base is 199.186 kV.

TABLE 2. Eight PMU rows of the executed measurement map. V is a bus voltage in p.u.; I is an oriented branch-terminal phasor. Gains and phase biases are applied to current channels. BUS5 retains the historical 5–6, terminal-5 hypothesis and its producer CT identity is unverified. BUS39 current phase is partially validated.

PMU	V bus	Edge	Branch	Terminal	Side	Polarity	Gain	Bias (°)	I unit	Provenance
2	2	2	2–3	3	dst	1	0.9581	-1.173	A	Terminal hypothesis; CT unverified
5	5	9	5–6	5	src	1	1.0361	0.312	A	BUS5 5–6 historical hypothesis; CT unverified
6	6	11	6–7	6	src	1	1.0077	1.495	A	Terminal hypothesis; CT unverified
10	10	17	10–11	11	dst	-1	1.0041	2.224	A	Terminal hypothesis; CT unverified
19	19	26	16–19	19	dst	1	1.0384	0.481	A	Terminal hypothesis; CT unverified
22	22	34	21–22	22	dst	1	1.0189	0.094	A	Terminal hypothesis; CT unverified
29	29	44	28–29	29	dst	-1	1.0316	-6.170	A	Terminal hypothesis; CT unverified
39	39	1	1–39	39	dst	-1	0.9298	30.053	A	BUS39 partial phase validation

TABLE 3. Fitted ExtraTrees comparators used in the held-out streams. Both were trained for this study and are distinct from the original competition models.

Comparator	Features	Windows	Trees
ExtraTrees	846	1, 5, 10 s	400
ExtraTrees short	1457	1, 5, 10 s; 3, 8, 15 reports	400

Frequency is computed as $60 + \dot{\theta}/(360)$ from the reported voltage angle, and ROCOF as $\ddot{\theta}/360$ using a causal 21-report quadratic estimate. The generator fit is robust and magnitude weighted; the predictor uses a fixed unweighted quadratic fit over the same history. Four warmup outputs are unavailable; the confirmatory evaluation uses a 10-s startup context and begins monitoring after 12 s. The 47 retained coordinates exclude the BUS39 current-angle coordinate. Noise multipliers 1 and 2 scale the standard deviation under the fixed stream. The stochastic model stores a cross-signal covariance field, but the executed report operator applies noise by signal family without that cross-family factor. These choices describe a simulated sensor configuration, not a verified eight-PMU CT/PT installation.

APPENDIX F LEARNED COMPARATOR LINEAGE AND DECISION CONTRACT

Both models were fitted on 32 physical parents from four earlier development campaigns; validation used 8 parents and 16 observed records from a fifth campaign. All of these were development data for the comparator construction. The fitted label set contains normal operation and all 18 catalog sources. The training target is the active source; after restoration, including physical recovery, its label is normal. This is not a source-excluded transfer test.

Each report selects the largest tree vote, and six consecutive reports of the same label drive the episode interface; there is no separate probability threshold. Tree votes are uncalibrated; the 300-report prefix and the start of monitoring at report 360 are fixed. Model hashes, training metadata and the observed class set are kept in the reproducibility package.

APPENDIX G COMPLETE CONFIRMATORY OUTCOMES AND DIAGNOSTIC AUDIT

The fixed test contains 171 physical trajectories and 342 observed records. Every method has an accounted output for every intended record; the solver-rejection and inference-failure ledgers are empty. The continuous, persistent-line and normal physical-parent counts are 18, 27 and 12. Three network conditions and two noise levels repeat these parents. Normal controls are shared between evaluation lanes, not doubled independent exposure. Tables below are generated directly from the stored summary, records, event matches and paired intervals.

A. TIMELY COVERAGE VERSUS CLASSIFICATION

The additional confusion view in Fig. 1 is a posthoc descriptive audit. It uses the original evaluator's temporal episode association and the first public diagnosis; it does not rematch events using source labels. A diagnosis emitted after physical OFF is counted as unavailable for this view. All diagnoses that are timely and matched in these four procedures have the correct family and exact source. The large load gap therefore reflects whether a diagnosis is available before OFF, not a measured advantage in conditional label accuracy among timely outputs. Unmatched alarms remain in the separate alarm tables. Normal records are not mixed into this event-level matrix.

B. MAINTAINING A SUSTAINED EVENT AFTER THE FIRST STOP

The first-stop score alone hides subsequent fragmentation. Although both physical banks and both learned comparators correctly identify all 162 persistent openings first, the learned comparators later return to an inactive label and issue additional alarms. Table 11 describes those already recorded outputs. It does not change the fixed first-stop outcome or extend Chen's one-stop contract into a recurrent monitor. Retained open episodes are appropriate when an intervention continues beyond the record.

C. MEASUREMENT INTEGRITY IS A SEPARATE EXECUTED COMPONENT

The availability interface correctly identifies all four missing-data intervals at the first received report that reveals the loss. This is a received-report clock, not zero transport or

TABLE 4. Continuous outcomes by network and noise stratum. Each cell is timely family/source decisions / intended events; unmatched event-stream alarms. Replays share physical parents.

Network	Noise	Physical bank	Nominal bank	ExtraTrees	ET short
Matched	1	108/108; 1	108/108; 1	56/108; 6	60/108; 5
Matched	2	108/108; 0	108/108; 0	54/108; 4	58/108; 0
In-bank	1	108/108; 2	108/108; 3	56/108; 5	60/108; 5
In-bank	2	108/108; 1	108/108; 2	54/108; 4	57/108; 0
Out-of-bank	1	106/108; 38	105/108; 47	56/108; 5	60/108; 6
Out-of-bank	2	106/108; 30	106/108; 39	54/108; 5	57/108; 0

TABLE 5. Continuous monitoring: unavailable intended records, normal-record alarm incidence, normal alarm counts and available normal exposure, false restorations and open episodes. Counts combine six repeated strata descriptively; normal parents are not independent across strata. Missing outputs remain intended failures.

Method	Unavailable	Normal incidence	Normal alarms	Exposure (h)	False OFF	Open at end
Physical 5-model	0	0/72	0	11.760	58	0
Physical nominal	0	0/72	0	11.760	68	0
ExtraTrees	0	0/72	0	11.760	29	0
ET short	0	0/72	0	11.760	16	0

TABLE 6. Conditional median diagnosis/restoration delays (s), kept within each paired stratum. These medians summarize matched emitted diagnoses/restoration messages with finite delay and must be read alongside omissions, unavailable records and censoring; – denotes no finite estimate. Logical delays do not measure computational throughput.

R/X condition	Noise	Physical 5-model	Physical nominal	ExtraTrees	ET short
Matched	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
Matched	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
In-bank	1	0.27/0.18	0.27/0.18	0.23/0.18	0.19/0.18
In-bank	2	0.27/0.18	0.27/0.18	0.30/0.18	0.20/0.18
Out-of-bank	1	0.33/0.18	0.33/0.18	0.23/0.18	0.19/0.18
Out-of-bank	2	0.26/0.18	0.33/0.18	0.30/0.18	0.20/0.18

TABLE 7. Persistent-line comparison on the common observations. Each cell gives correct first source / intended openings; pre-change stops / right-censored openings; mean restricted correct delay (s). Wrong first sources remain in the denominator. Restricted delay retains the horizon penalty for failures and censoring defined by the fixed evaluator.

R/X condition	Noise	Physical 5-model	Physical nominal	Chen *et al.*
Matched	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Matched	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
In-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.04
In-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	13/27; 11/0; 51.05
Out-of-bank	1	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.58
Out-of-bank	2	27/27; 0/0; 0.18	27/27; 0/0; 0.18	12/27; 11/0; 54.59

computation latency. Bad-data labels come from the inherited window classifier. Neither missingness nor corruption proves malicious intent. The composition preserves physical-source and affected-PMU identities but does not run a joint physical/cyber Bayesian filter; the physical bank is unavailable on the retrospective replay of the competition reference record under its startup noise-scale contract.

TABLE 8. **Persistent-line coverage and stopping outcomes across six repeated strata, including both fixed ExtraTrees comparators. Normal controls are the same controls used in the continuous table, not additional exposure. Chen *et al.* normal exposure ends at its first stop; the other methods monitor the full available record.**

Method	Unavailable	Correct first	Pre-change	Censored	Normal incidence	Exposure (h)
Physical 5-model	0	162/162	0	0	0/72	11.760
Physical nominal	0	162/162	0	0	0/72	11.760
ExtraTrees	0	162/162	0	0	0/72	11.760
ET short	0	162/162	0	0	0/72	11.760
Chen *et al.*	0	74/162	66	0	72/72	0.701

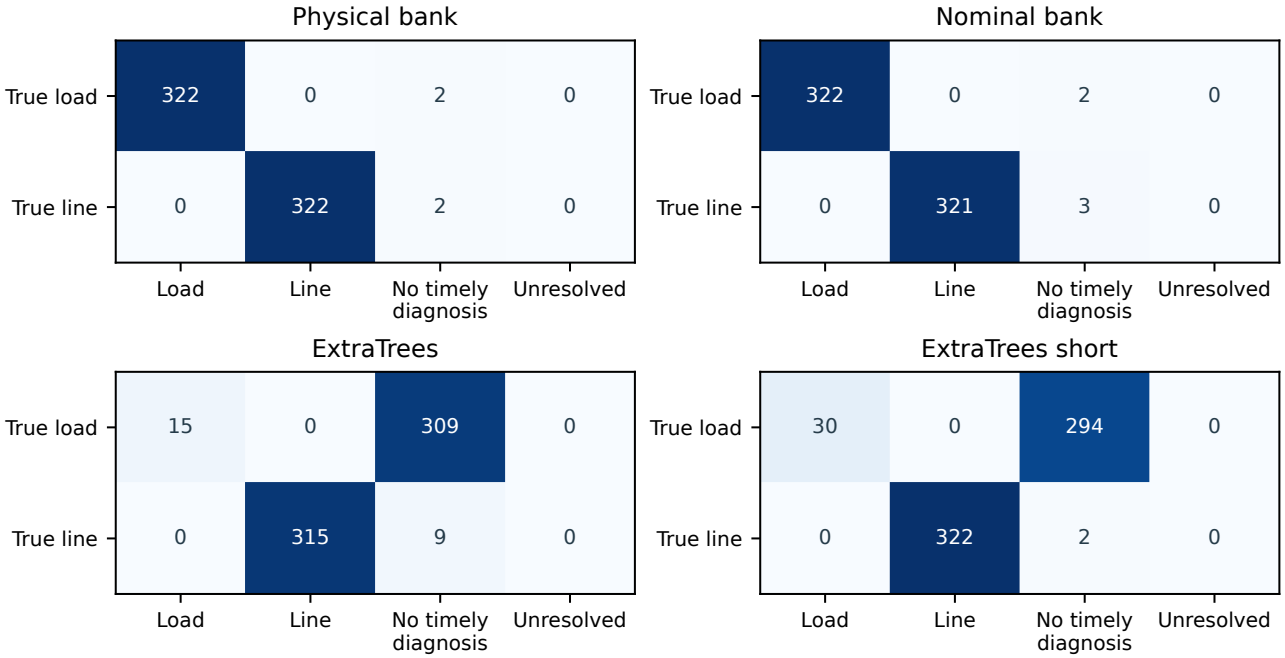


FIGURE 1. **Posthoc confusion audit of the fixed continuous test, with 324 realizations per true family. No timely diagnosis includes omitted and late diagnoses. Counts reuse the registered episode matches; all 432 prediction JSON files used were checked against the stored prediction hashes. These panels do not include unmatched alarms or constitute a new predeclared endpoint.**

TABLE 9. **Timely decisions by source in the continuous test. These are correlated realizations of the registered physical schedules, not source-excluded generalization.**

Family / asset	Physical bank	Nominal bank	ExtraTrees	ET short
Line / LINE14-15	36/36	36/36	27/36	34/36
Line / LINE16-17	36/36	36/36	36/36	36/36
Line / LINE17-18	48/48	48/48	48/48	48/48
Line / LINE23-24	30/30	30/30	30/30	30/30
Line / LINE26-27	24/24	24/24	24/24	24/24
Line / LINE26-29	24/24	23/24	24/24	24/24
Line / LINE3-4	42/42	40/42	42/42	42/42
Line / LINE4-14	46/48	48/48	48/48	48/48
Line / LINE9-39	36/36	36/36	36/36	36/36
Load / BUS15	42/42	42/42	6/42	6/42
Load / BUS16	42/42	42/42	0/42	3/42
Load / BUS20	12/12	12/12	6/12	12/12
Load / BUS24	42/42	42/42	0/42	0/42
Load / BUS26	18/18	18/18	0/18	0/18
Load / BUS3	42/42	42/42	3/42	3/42
Load / BUS31	40/42	40/42	0/42	0/42
Load / BUS4	36/36	36/36	0/36	6/36
Load / BUS7	48/48	48/48	0/48	0/48

TABLE 10. Paired physical-bank minus learned-comparator differences. Timely attribution is in percentage points; extras are per monitored event-record hour, excluding 12 s of startup from each 120-s record. The registered rate differences and 95-percent parent-bootstrap bounds are exactly rescaled by 120/108; original counts, matches and intervals remain stored.

Network	Noise	Method	Reference	Timely difference	Extra-rate difference
Matched	1	Physical bank	ExtraTrees short	44.44 [40.48, 47.92]	-7.41 [-16.67, 0.00]
Matched	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-9.26 [-20.37, 0.00]
Matched	1	Nominal bank	ExtraTrees short	44.44 [40.74, 48.04]	-7.41 [-14.81, 0.00]
Matched	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-9.26 [-20.37, 0.00]
Matched	2	Physical bank	ExtraTrees short	46.30 [42.75, 49.12]	0.00 [0.00, 0.00]
Matched	2	Physical bank	ExtraTrees	50.00 [46.30, 53.70]	-7.41 [-14.81, -1.85]
Matched	2	Nominal bank	ExtraTrees short	46.30 [42.86, 49.12]	0.00 [0.00, 0.00]
Matched	2	Nominal bank	ExtraTrees	50.00 [46.38, 53.70]	-7.41 [-14.81, -1.85]
In-bank	1	Physical bank	ExtraTrees short	44.44 [40.62, 47.92]	-5.56 [-14.81, 5.56]
In-bank	1	Physical bank	ExtraTrees	48.15 [44.44, 51.67]	-5.56 [-14.81, 5.56]
In-bank	1	Nominal bank	ExtraTrees short	44.44 [40.62, 47.92]	-3.70 [-12.96, 7.41]
In-bank	1	Nominal bank	ExtraTrees	48.15 [44.44, 51.67]	-3.70 [-12.96, 5.56]
In-bank	2	Physical bank	ExtraTrees short	47.22 [43.18, 50.98]	1.85 [0.00, 5.56]
In-bank	2	Physical bank	ExtraTrees	50.00 [46.49, 53.70]	-5.56 [-11.11, 0.00]
In-bank	2	Nominal bank	ExtraTrees short	47.22 [43.33, 51.04]	3.70 [0.00, 9.26]
In-bank	2	Nominal bank	ExtraTrees	50.00 [46.30, 53.70]	-3.70 [-11.11, 3.70]
Out-of-bank	1	Physical bank	ExtraTrees short	42.59 [38.60, 46.43]	59.26 [40.74, 79.63]
Out-of-bank	1	Physical bank	ExtraTrees	46.30 [42.11, 50.00]	61.11 [40.74, 81.48]
Out-of-bank	1	Nominal bank	ExtraTrees short	41.67 [36.11, 46.67]	75.93 [51.85, 100.00]
Out-of-bank	1	Nominal bank	ExtraTrees	45.37 [41.11, 50.00]	77.78 [50.00, 105.56]
Out-of-bank	2	Physical bank	ExtraTrees short	45.37 [41.11, 50.00]	55.56 [38.89, 72.22]
Out-of-bank	2	Physical bank	ExtraTrees	48.15 [43.59, 52.50]	46.30 [27.78, 66.67]
Out-of-bank	2	Nominal bank	ExtraTrees short	45.37 [41.11, 50.00]	72.22 [46.30, 98.15]
Out-of-bank	2	Nominal bank	ExtraTrees	48.15 [43.75, 52.56]	62.96 [35.19, 88.89]

TABLE 11. Descriptive persistence audit after first detection on sustained line openings. All four procedures first identify 162/162 lines. Extras are later unmatched alarms; unmatched OFF counts only restoration messages on unmatched episodes. Open-at-end messages are expected for sustained events and are not automatically errors. False-negative reports include detection delay and later inactive labels. Chen is a first-stop method and is excluded from this recurrent-monitoring diagnostic.

Method	Openings	Extras	Unmatched OFF	Open at end	FN reports
Physical bank	162	0	0	162	810
Nominal bank	162	0	0	162	810
ExtraTrees	162	164	164	16	335374
ExtraTrees short	162	179	178	61	182600

TABLE 12. Retrospective measurement-integrity components over 180 windows. Hits require family and PMU identity. Missingness is observed availability; corrupted-data identification is inherited from the window classifier. These are not counts of attributed cyberattacks.

Component	Positive-window hits	FP	FN
Missing	14/14	0	0
Corrupt	1/2	2	1

APPENDIX H CONTROLLED COMPONENT ABLATION

The fixed protocol and all output tables are part of the reproducibility package. Each of four arms accounts for all 24 intended records (18 continuous, three normal and three persistent): the evaluation audit records 96/96 available outputs and zero failures. The trial is posthoc and no score threshold is calibrated to these outcomes. Each normal control is a 600-s record with 588 s of evaluated exposure after startup, for 1,764 s total.

a: Exposure convention for the ablation alarm rate.

The ablation table reports extras per event-hour with the complete 120-s event record, including startup, as denominator. Its continuous exposure is $18 \times 120 = 2160$ s. The main article's confirmatory alarm-rate figure instead uses monitored exposure after the 12-s startup. Under that convention the ablation exposure would be $18 \times 108 = 1944$ s and each reported rate difference and interval bound would be multiplied by $120/108$. Counts, event matches and the stored table are unchanged; these two rate conventions must not be compared without this conversion.

APPENDIX I EXECUTABLE SINGLE-STATE COMPARISON

This control uses the same conditional nominal plant and measurement operators as the nominal history bank, but retains only one committed mode and its augmented conditional state. Temporary alternatives live for at most 19 reports. The five-model history bank additionally mixes resistance and reactance alternatives. Proposals begin at report index 300 and publication at index 360, with zero-based indices. Each available report after proposal startup starts two subframe transition candidates for every admitted source when inactive, or two OFF candidates for the committed source when active. Every candidate uses the covariance regime chosen from preceding reports; the current sample cannot select its own predictive covariance.

For a load ON alternative, write the innovation as $e_k - as_k$. The affine conditional state is $m_k + ad_k$. Accumulate over its contiguous suffix

$$G = \sum_k s_k^\top S_k^{-1} s_k, \tag{21}$$

$$U = \sum_k s_k^\top S_k^{-1} e_k, \tag{22}$$

$$C = -\frac{1}{2} \sum_k \left(e_k^\top S_k^{-1} e_k + \log |S_k| + p \log(2\pi) \right). \tag{23}$$

Then $\widehat{a} = \mathrm{clip}(U/G, -0.2, 0.2)$ when $G > 0$, with $\widehat{a} = 0$ for a flat objective. The profile contrast is

$$\Lambda = C + \widehat{a}U - \tfrac{1}{2}\widehat{a}^2 G - C_{ref}. \tag{24}$$

Here C_{ref} is the sum of the committed predictor's pre-update conditional log densities on exactly the same suffix. A line uses its discrete switched operator, without fitting a load amplitude. The reference uses its preceding severity estimate for prediction and can refine that estimate after receiving the new report. A candidate never changes the reference state before commitment. A load OFF alternative holds its birth amplitude fixed rather than re-profiling it.

ON scores subtract $\log(18 \times 2 \times 19)$; OFF scores subtract $\log(2 \times 19)$. These fixed search penalties are heuristics. Maximizing over overlapping suffixes and bounded amplitudes does not produce a calibrated posterior or a sequential false-alarm guarantee. For the history banks, ON uses the log odds of $M(c)$ and OFF the log odds of $M_{cause}(c)P(OFF \mid c)$.

The common clock requires the same winning ON source to exceed $\log 99$ for six consecutive monitored reports. It emits alarm and source diagnosis together. Six qualifying OFF reports for that source close the episode. A clock-induced commitment selects the candidate's current conditional state. Its estimate may change through hypothesis selection; it is never replaced with equilibrium or a zero state. A missing report propagates the retained mode without measurement correction, clears the short alternatives and contributes no confirmation evidence. The development comparison contains no such post-startup missing observations.

TABLE 13. All outcomes, including restoration messages on unmatched episodes and episodes open at the record end.

method	lane	available	events	timely	extras	false restorations	open eof
Marginal + carry	continuous	18	108	108	1	1	0
Marginal + carry	normal	3	0	0	0	0	0
Marginal + carry	persistent line	3	3	3	0	0	3
Marginal + reset	continuous	18	108	75	166	130	11
Marginal + reset	normal	3	0	0	0	0	0
Marginal + reset	persistent line	3	3	3	0	0	3
Profile + carry	continuous	18	108	108	1	1	0
Profile + carry	normal	3	0	0	0	0	0
Profile + carry	persistent line	3	3	3	0	0	3
Profile + reset	continuous	18	108	75	166	130	11
Profile + reset	normal	3	0	0	0	0	0
Profile + reset	persistent line	3	3	3	0	0	3

TABLE 14. Paired descriptive differences with 95-percent physical-parent bootstrap intervals; no multiplicity-adjusted superiority claim.

method	reference	metric	difference	lower95	upper95
Marginal + carry	Profile + carry	timely fraction	0.000	0.000	0.000
Marginal + carry	Profile + carry	extras per event-hour	0.000	0.000	0.000
Marginal + carry	Marginal + reset	timely fraction	0.306	0.241	0.380
Marginal + carry	Marginal + reset	extras per event-hour	-275.000	-325.000	-223.333
Marginal + reset	Profile + reset	timely fraction	0.000	0.000	0.000
Marginal + reset	Profile + reset	extras per event-hour	0.000	0.000	0.000
Profile + carry	Profile + reset	timely fraction	0.306	0.241	0.380
Profile + carry	Profile + reset	extras per event-hour	-275.000	-325.000	-223.333

TABLE 15. Open-development comparison with the same six-report publication clock. All 54 event replays are retained. Threshold score semantics differ; this does not establish equal false-alarm operating points. Missing closure includes missed events.

Procedure	Timely	Extras	Early OFF	No OFF
Single state, short GLR	37/54	37	0	4
Nominal history bank	44/54	6	0	10
Five-model history bank	51/54	3	0	3

TABLE 16. What the causal component resolves in the second development campaign. Load and line columns require the correct family and source emitted before OFF. The 54 realizations repeat 18 interventions across three R/X conditions; they are not 54 independent trials. Normal exposure is only 204 s per method.

Method	Load timely	Line timely	Extras	Normal alarms
Physical (5 models)	26/27	27/27	4	0
Physical nominal	27/27	27/27	7	0
ExtraTrees ref.	0/27	27/27	3	0
ET short	0/27	27/27	3	0
Onset witness	25/27	27/27	0	0

After OFF, the preceding discrepancy regime persists until 60 consecutive available, inferred-inactive reports meet the inherited quiet criterion. Its innovation is computed with the normal operator from the preceding committed state. A new active episode or missing report clears that quiet run. Retrospective source labels and event times never enter these steps.

The comparison has one fixed publication clock but different evidence semantics, search and severity treatment. It therefore compares complete procedures. It does not establish a matched false-alarm operating point, nor isolate a causal effect of long-history averaging. The source tree, execution hashes, every prediction and the successor event metrics are preserved in the experiment manifest.

APPENDIX J DEVELOPMENT STREAMS AND ADMISSION CONTROL

A. TIMELY ATTRIBUTION AND EXTRA ALARMS MEASURE DIFFERENT OUTCOMES

In the second development campaign, both ExtraTrees adapters and both physical references identify all tested lines in time (Table 16). The difference is in loads: the five-model bank obtains 26/27 timely decisions, nominal 27/27, and these ExtraTrees adapters 0/27. This tests transfer of the fixed adapters; it neither rules out supervised load detection nor reproduces the historical multi-class hackathon score.

The five-model bank detects and eventually attributes 54/54 realizations, but only 53 diagnoses precede physical OFF. Nominal has 54 timely diagnoses, with 7 extras versus 4. Detection, eventual attribution and timely attribution therefore give different answers. The configured five-model procedure yields one fewer timely diagnosis and three fewer extra alarms than nominal; this comparison does not isolate model uncertainty.

B. RECOVERY AND A NEW EVENT MUST REMAIN COMPETING EXPLANATIONS

Figure 3 shows what the counts omit. The physical state continues evolving after OFF. Retained cause/recovery histories explain that trajectory without resetting the network; a later source must compete against both no-new-event recovery and other interventions.

The out-of-bank BUS31 case shows a remaining timing

failure: the source is recovered eventually but too late. The network mixture changes how much mass is assigned to that source, and the mass-increment identity in the main article separates that revision from within-model evidence. This decomposition distinguishes changing within-model evidence from changing network weights; it is an interpretation of the delayed decision.

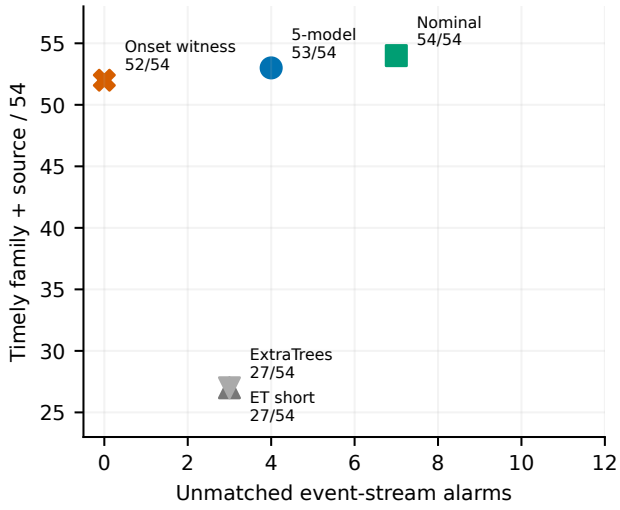


FIGURE 2. The decision that an accuracy-only table hides. Development outcomes trade timely attribution against unmatched event-stream alarms. The onset-witness ablation removes all extras but loses a real BUS16 intervention; it is retained as negative evidence. Labels use 54 paired realizations of 18 interventions and do not imply 54 independent trials.

The onset-witness ablation obtains zero extras but only 52/54 timely attributions: weak early evidence rejects a real BUS16 load. It inherits the physical states and masses, so suppressing the alarm does not repair the underlying explanation. This negative result left the five-model and nominal banks unchanged for the independent evaluation.

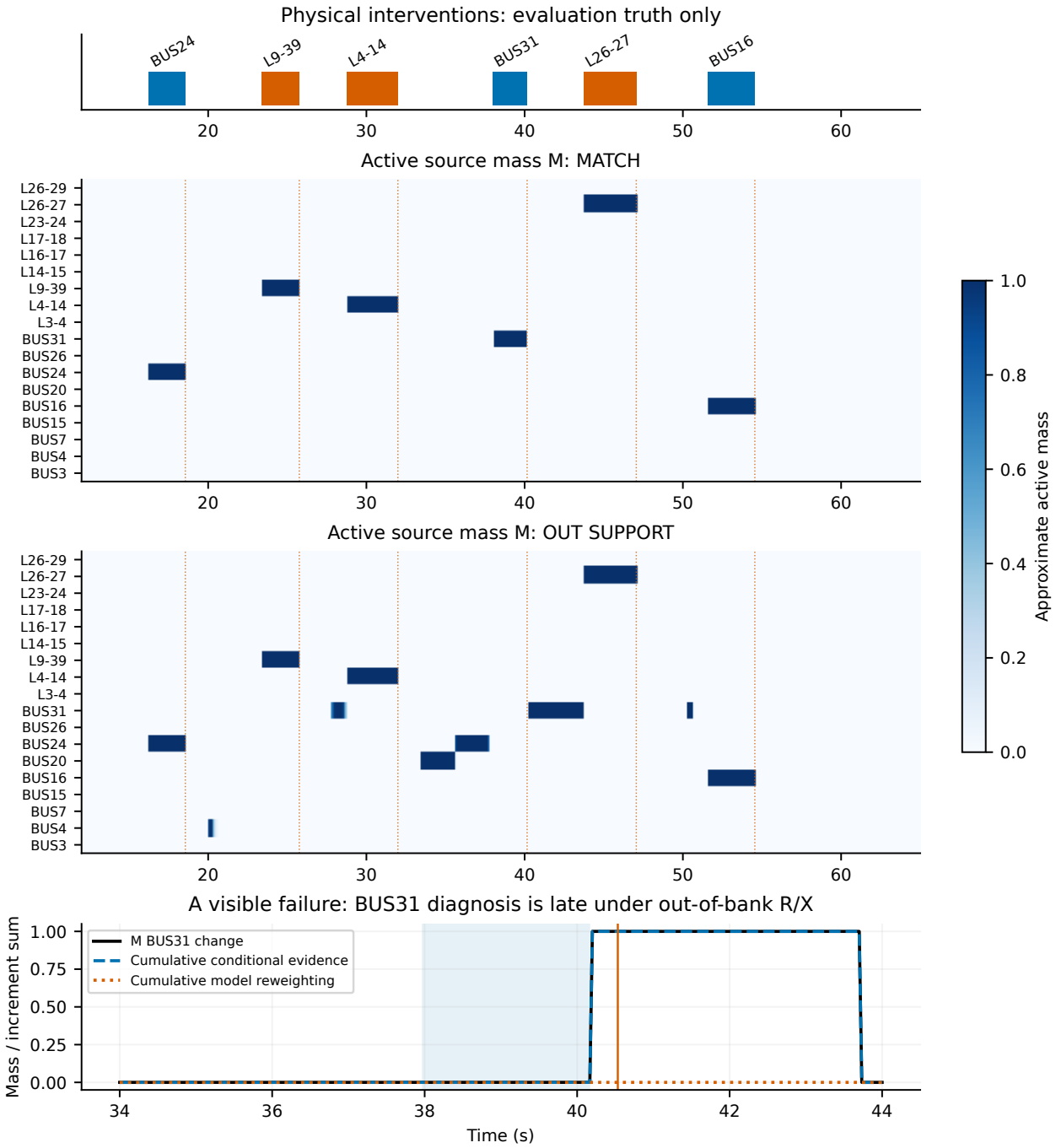


FIGURE 3. **From a continuous sequence to source attribution. Top: reserved physical ON–OFF intervals in one development parent. Middle: active masses for nominal-network observations (MATCH) and mixed R/X changes outside the bank (OUT SUPPORT). Orange dotted lines mark true OFF for evaluation only. Bottom: BUS31 in the latter record; the cumulative terms of $\acute{M}$, rebased at 34 s, sum to the change in active mass. Diagnosis arrives after OFF. The figure shows an eventual correct source that fails timely attribution; the color scale represents approximate mass, not calibrated probability.**

APPENDIX K DETECTABILITY COMPUTATION AND NOISE-STRESS PROTOCOL

a: Operators.

The information analysis of the main article uses the nominal bank built by the evaluated implementation for one held-out calibration record of a later campaign whose runtime is byte-identical to the independent evaluation runtime. The independent-evaluation banks had been evicted by the storage-retention policy; the nominal operators differ between records only through the prefix-estimated operating factors, which remain within 2×10^{-5} of unity.

b: Signatures.

For each source i, the noise-free switched predictor is propagated from a zero pre-event deviation with the ON operator at half-phase 0 followed by the stay operator (loads at severity 0.1). Each candidate filter $j \in \{0, \dots, 18\}$ is run on these reports with the quiet-regime gains K_o and inverse innovation covariances S_o^{-1} of the implementation. The cumulative sums $\sum e^\top S^{-1} e$, $\sum s^\top S^{-1} e$ and $\sum s^\top S^{-1} s$ give the signature information of the main article with the competitor's severity profiled on $[-0.2, 0.2]$. As a check, the true-history filter at the true severity leaves a residual energy below 10^{-8}. The detection threshold uses the birth hazard $1 - \exp[-(1/60)/30]$ shared equally by the 18 cells.

c: Reset cost.

For every intervention of the controlled study, the noise-free truth is propagated through ON, the active interval, OFF and the gap until the next onset. At the inferred closure time of the carry arm, the first 114 coordinates of the true augmented state are zeroed, and the no-intervention and all birth filters are run on the remaining truth in the recovery covariance regime. The largest profiled gain of a birth over the no-intervention history is compared with the alarm evidence $\lambda^\star$. Harm in the reset arm is an unmatched alarm between the OFF of the intervention and the next onset, or a lost timely attribution of the next intervention.

d: Noise stress.

Observations are regenerated from the stored deterministic trajectories with the simulator's own noise stream and seed: $\varepsilon = c + \alpha(v - c)$ for multiplier α, where v is the stream and c its normalization center, exactly as the campaign exporter. The $\alpha = 1$ observations reproduce the original files to 6×10^{-10} and the inference reproduces all 18 published messages of the parity record. The TVE scenarios add independent zero-mean Gaussian phasor noise per PMU and report, with relative magnitude and angle (radian) standard deviations of 5×10^{-4} or 10^{-3}; the added voltage-angle noise is passed through the simulator's causal 21-report quadratic frequency and ROCOF estimator. The nominal bank is rebuilt from the stored derivative matrices with the noise scale estimated from the noisy prefix. The operating point remains at its $\alpha = 1$ estimate because derivatives exist only there. Estimated angle noise scales of the TVE scenarios are about 48 times the calibrated scale, outside the implemented support $[0.1, 10]$; for this test only the support was widened to $[0.1, 1000]$. Two jobs ran concurrently with another campaign on the same desktop CPU, so the reported processing rates are conservative.

e: Reproduction.

Every number and figure of the analysis is regenerated from the stored operators and outcomes by the reproducibility package, which also contains the noise-stress generator.

APPENDIX L OPERATING-POINT SIGNATURE EXPERIMENT

A. REFERENCE CORRECTION AND RESPONSE CORRECTION

The stream bank already builds local predictors at estimated operating points. A separate experiment tests why changing a reference alone may be insufficient. It compares three response approximations with identical observations, channels, timing, amplitude prior and inference horizon. It is a controlled test of a mixed derivative, not another version of the stream method.

Let η denote a scalar operating change and a the signed load factor. Central differences of complete nonlinear DAE trajectories give

$$D_i(\eta) \simeq \frac{y_i(\eta, h) - y_i(\eta, -h)}{2h}, \qquad E_i \simeq \frac{D_i(d) - D_i(-d)}{2d}. \tag{25}$$

The event follows the same ON/OFF schedule in every derivative trajectory. The fixed Q_i^0 is the coefficient of a^2, obtained from the centered second difference divided by $2h^2$. After removing the fixed common angle drift, the three predicted histories are

$$\begin{aligned} \mu_i^{\mathrm{N}}(a) &= y_0 + aD_i^0 + a^2 Q_i^0, \\ \mu_i^{\mathrm{R}}(a) &= \mu_i^{\mathrm{N}}(a) + \widehat{b}, \\ \mu_i^{\mathrm{E}}(a) &= \mu_i^{\mathrm{R}}(a) + a\widehat{\eta} E_i. \end{aligned} \tag{26}$$

Both corrected variants use exactly the same prefix offset $\widehat{b}$. This comparator corrects the measured PMU reference; it is not the separate nonlinear AC hidden-state reconstruction algorithm in the historical study. Only the last changes the event signature. The first two consequently have identical incremental response models even when their absolute fits differ. Context $\widehat{\eta}$ is estimated by generalized least squares from the pre-event PMUs and the nominal context direction. No test context, source or severity enters inference.

B. TEST CASES AND DECISION CONTRACT

The research plant supplies all trajectories. The scalar η scales PQ operating injections and active PV dispatch; feasible initialization can change permitted operating setpoints, but not machine constants, line impedances, ZIP fractions or the PMU model. Central event/context steps are 0.005 and 0.01, with step-halving checks. Dictionary construction

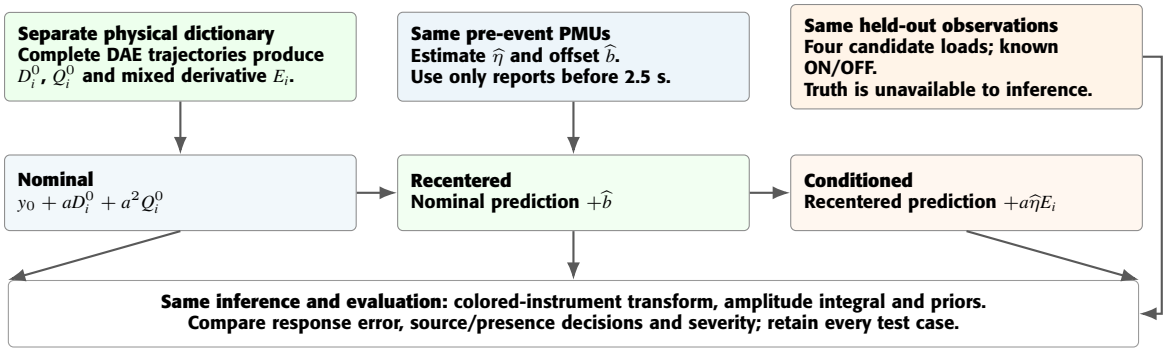


FIGURE 4. Executed signature ablation, separate from the causal stream algorithm in the main article. The three variants share the data and decision calculation; only reference and event-response terms differ.

uses 57 trajectories. The test contains 85 physical cases: 80 load interventions and 5 controls at operating changes $-3\%, -2\%, 0, +2\%, +3\%$. Each load changes P/Q together by $\pm 0.1\%$ or $\pm 1\%$ at BUS3, BUS7, BUS16 or BUS24. None belongs to the prescribed voltage-measurement PMU set; the separate oriented current-terminal assignments remain fixed.

Each case is one ten-second integration, with ON at 3 s, OFF at 6 s and no state reset. The context fit ends at 2.5 s. Three fixed instrumental noise seeds produce 255 paired records. The nonlinear PMU frequency/ROCOF operator runs before its stored instrumental noise is added. Both corrected methods retain the same measured prefix, so an advantage cannot arise from supplying one method a better initial reference.

This experiment supplies the load family and ON/OFF times. It estimates source and continuous severity using the same bounded Gaussian prior and amplitude-integrated likelihood in all three variants. The no-event hypothesis is retained. The fixed colored-instrument transform is common; plug-in prefix estimates and clipped noise still limit probability calibration. Thus these decisions are not unknown-onset detector scores and normal-window decisions are not continuous false-alarm rates.

C. RESPONSE ERROR AND SOURCE DECISIONS

Relative to recentering alone, the mixed term reduces mean incremental response error: the paired difference is -1.15 percentage points (95% interval [-1.31, -1]). The corresponding joint-decision difference is 0 percentage points ([0, 0]), which does not establish a directional improvement in joint source decisions. The recentered and conditioned source decisions are already at the test ceiling. Median absolute relative severity error, evaluated at the generating source, changes from 1.2% to 0.329%. Its mean paired change is -0.738 percentage points ([-0.898, -0.582]). Both variants select the correct source in every tested event record, so this severity comparison does not discard source errors. Their source masses numerically saturate at one; zero recorded log-loss is not evidence of calibrated uncertainty.

The gain is not uniform across all comparisons: at the nominal operating point, estimating an unnecessary reference/context can worsen severity relative to keeping the correct fixed reference. Figure 5 retains that case rather than reporting only shifted operation.

The derivative stability checks change E_i by 0.283–0.427% when the context step is halved. Halving the event step in the two checked directions changes it by at most 0.00977%. These are local numerical checks, not a global smoothness or identifiability certificate. Percentages of event severity refer to the P/Q input factors; actual load power also responds to voltage under the fixed impedance law.

Paired intervals resample physical test cases, keeping their noise seeds and methods together. The same three noise seeds are reused across cases; these intervals describe case variation conditional on those common noise realizations, not uncertainty over a noise population or transfer to arbitrary sources or plants. The step-halving checks and all case outcomes remain in the fixed artifact package. Better fidelity, if observed, is not sufficient to establish calibrated source probabilities, all-family diagnosis or a benefit in the unchanged causal campaign.

APPENDIX M RECOVERY GEOMETRY AND SOURCE SEPARATION

A. RESPONSE FIT, SEVERITY AND SOURCE IDENTITY

A retained physical history predicts the response still present after restoration and defines the state from which a new intervention begins. For the exact observation map Ψ and the same pre-intervention history h, the incremental response is

$$s_{e|h}(t) = \Psi_t(h \oplus e) - \Psi_t(h). \tag{27}$$

The two paths share their initial physical and instrumental states. In an additive linear time-invariant system, superposition makes this increment independent of the prior state. Nonlinear dynamics can change its shape, but do not guarantee greater source separation. Preserving recovery is a model-consistency requirement, not a claim that an earlier disturbance universally helps localization. The stream implementation approximates these paths with its declared switched predictors; it does not solve the full nonlinear DAE for each hypothesis.

TABLE 17. Controlled signature test, separate from causal detection. Response error is the median whitened incremental error at the generating source/severity over 80 physical cases. Joint correctness includes presence and source on 240 noisy event records; source-only ranking conditions on a load. Severity error is median absolute relative error at the generating source. Normal selections use 15 fixed windows, not normal-hours exposure.

Model	Response error (%)	Joint correct	Source-only correct	Normal event selections	Severity error (%)
Nominal	1.25	114/240	114/240	12/15	3.45
Recentered	1.25	240/240	240/240	0/15	1.2
Conditioned	0.0561	240/240	240/240	0/15	0.329

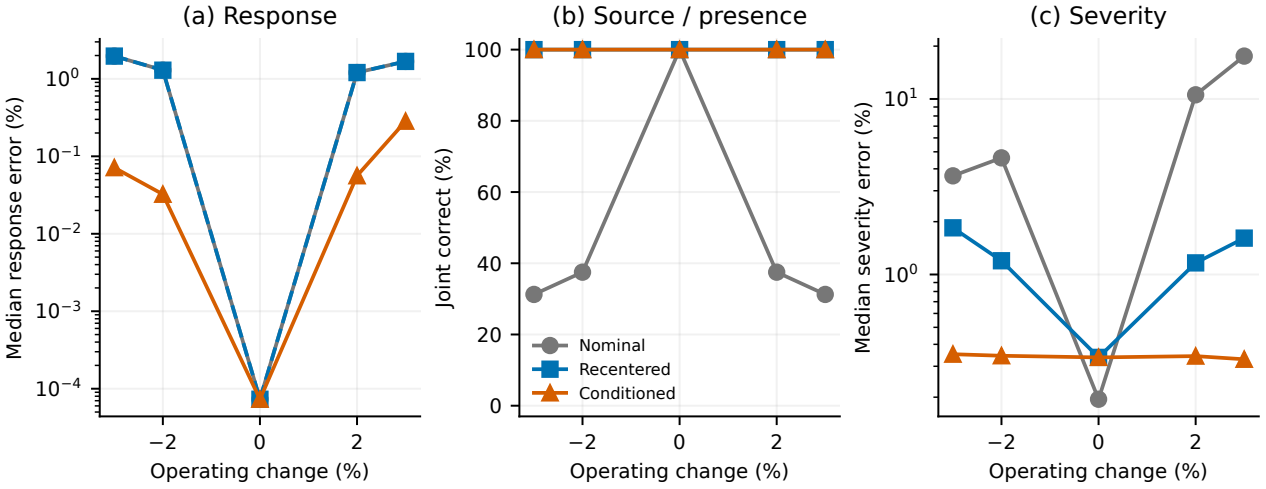


FIGURE 5. Separate effects of recentering and conditioning. (a) Incremental response error in the fixed physical test cases, evaluated at the generating source and severity to isolate approximation error. (b) Correct source and presence decisions on noisy observations, after marginalizing severity. (c) Relative severity error evaluated at the generating source. The panels measure different outcomes. Noise realizations repeat physical cases and are not counted as additional independent trajectories. Nominal and recentered incremental predictions coincide.

A better physical prediction need not produce a better location. Competing sources may remain similar after their severities and initial conditions are fitted. For a response s_h at specified severity and context, the distance to an admitted competitor is

$$d_h^2 = \min_{j \neq h} \min_{\beta_j \in \mathcal{A}_j} \|s_h - s_j(\beta_j)\|^2_{\Sigma^{-1}}. \tag{28}$$

The set $\mathcal{A}_j$ fixes allowed signs, severities and nuisance coordinates. Small distance makes source discrimination noise-sensitive; it is not a universal error formula or a calibrated decision certificate [2], [3].

Three outcomes remain separate: prediction error at a specified physical source, ranking of competing sources, and uncertainty calibration. A response improvement establishes the first. It establishes the second only if source decisions improve under the same observations, horizon and decision rule. Neither establishes the third automatically. A nonlinear response comparison is needed before treating a surrogate's residual confusion as an unavoidable physical limit.

The separately compiled Historical Technical Notes retain the earlier state-reconstruction and simultaneous-load studies, with their separate observation contracts and prior-normalization qualification. Their known-onset scores are not combined with the causal results here. In particular, recovering hidden voltages does not certify the source that moved them.

To first order, a component of the context derivative parallel to a nominal source signature can be absorbed by fitting a different severity. Correct source ranking can therefore coexist with biased intensity. Components orthogonal to that signature cannot be removed by this rescaling. This explains why source accuracy can reach a test ceiling while response and severity errors still distinguish the models; it is a standard projection argument, not a new identifiability theorem.

B. DOES RECOVERY INCREASE SOURCE SEPARATION?

A separate physical control tests this possibility without changing the stream method. We solve 38 complete nonlinear trajectories of 8 s with 241 PMU reports each. A prior 8% load change at bus 7 or 24 is applied at 2 s and restored at 3 s. A weak 0.06% load change follows after 0.5 or 2 s. The candidate pairs are 18/27, 16/21 and 3/21, and the observation horizon is 1.5 s. Controls and prior-severity differences account for the remaining trajectories. Within each matched history, the secondary-event and no-secondary-event trajectories share the same prefix. Physical and instrumental states are retained through restoration.

This is an oracle component test: the prior trajectory, secondary onset, load family and candidate pair are supplied.

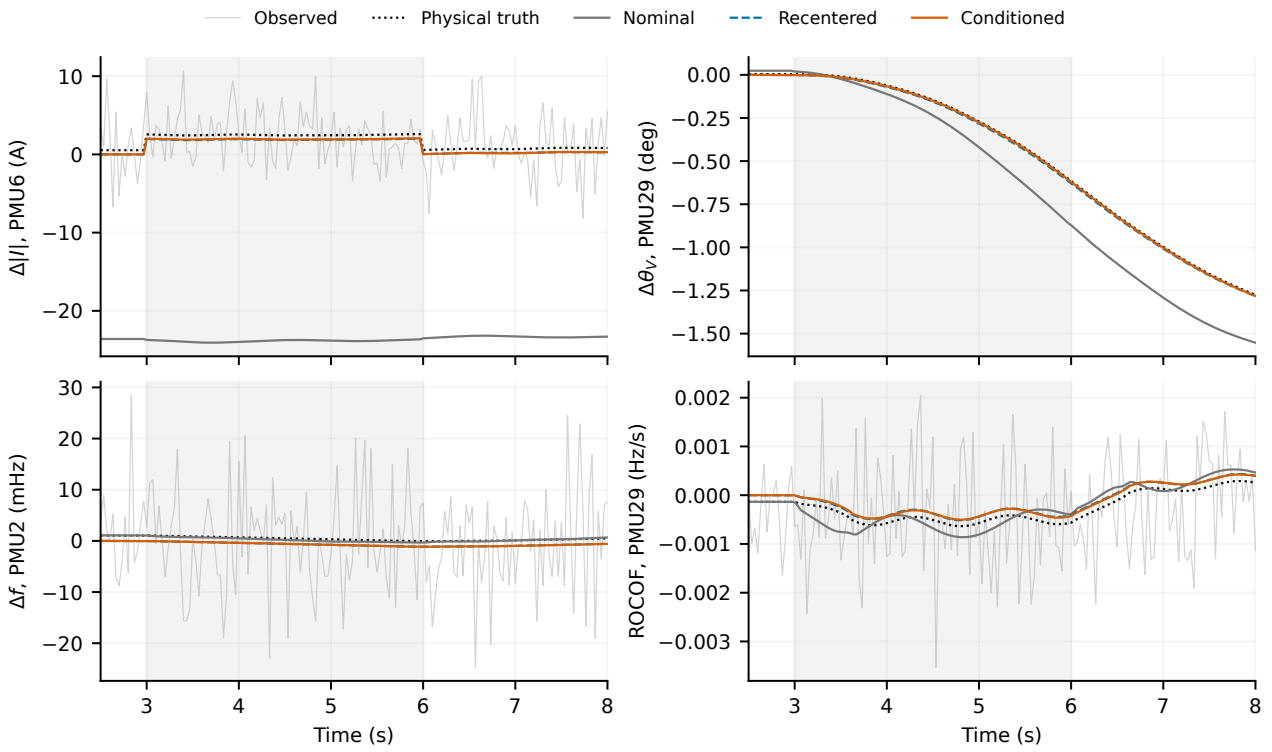


FIGURE 6. **Fixed illustrative test: BUS7, $\eta = +3\%$, load factor +1%, seed 390502. Gray traces are the delivered PMU observations. Colored curves use each method's selected source and estimated severity; a no-event selection keeps its baseline. The dotted black curve is physical truth, used only for illustration. The plotted channels and case were fixed without selecting the largest improvement. Shading marks physical ON/OFF for evaluation.**

Finite-amplitude differences of nonlinear trajectories provide incremental responses; these are secants, not newly derived analytical tangents. The 47-coordinate measurement operator is followed by its fixed innovations transform. For a geometric control, we profile nonnegative competing amplitudes. Adding a rank-one covariance from uncertainty of 0.01 in the previous load fraction tests sensitivity to a nuisance direction; it is not a causal posterior over the full state.

The median ratio is 1.0006 with known history and 0.8532 with that nuisance direction; none of the 12 latter ratios exceeds one (Fig. 7). An illustrative binary likelihood comparison, using a signed Gaussian amplitude prior of standard deviation 0.0015 and 300 common Gaussian innovation draws per generating source, gives the same mean accuracy, 98.35%, for equilibrium and conditioned signatures. Two cells worsen and one improves. This likelihood experiment integrates signed amplitudes, whereas the distance profiles nonnegative amplitudes; the distance is not asserted to bound its error. The noise assumption is an innovations-domain control, not validation of the complete instrumental generator. These correlated, known-onset comparisons establish neither an improvement of the stream method nor general equivalence. They provide no support for claiming that recovery universally creates useful source information.

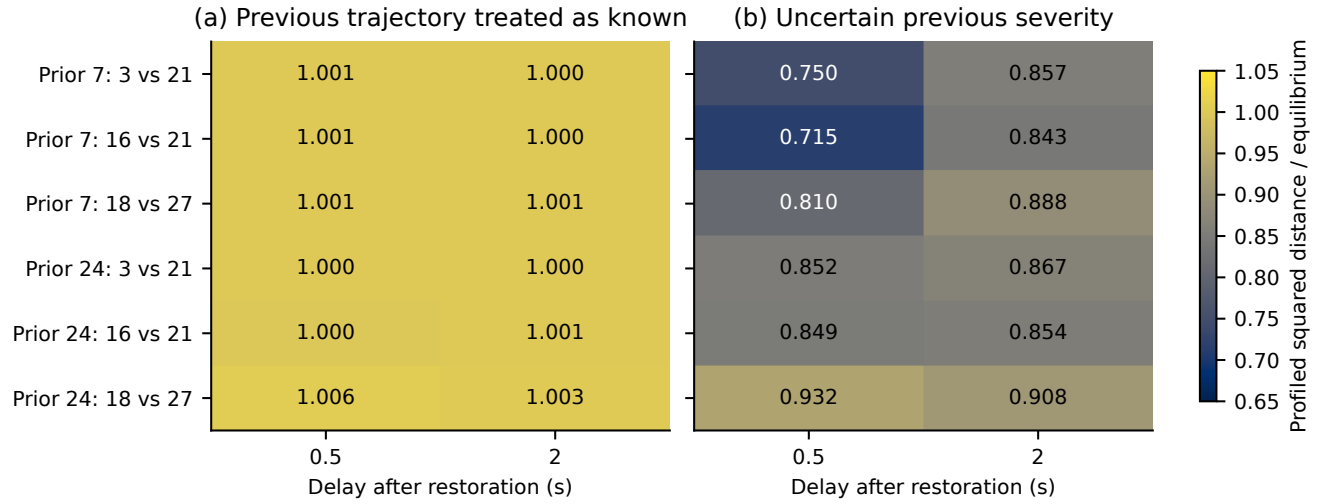


FIGURE 7. **Full nonlinear physical control, not a performance comparison of the stream method. Each cell is the profiled squared separation during recovery divided by its equilibrium value. The same source pairs are evaluated with the previous trajectory treated as known (left) and with one uncertain previous-severity direction (right). Values below one mean less separation. Adding nuisance covariance reduces precision at fixed signatures by construction; the left panel separately shows the small change in physical response geometry. The 12 cells share four recovery histories.**

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